

ON THE EVALUATION OF CERTAIN SERIES AT A GOLDEN RATIO-BASED CONSTANT

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ABSTRACT. This paper introduces and analyzes a specific complex constant, denoted Λ_{G1} , which is uniquely derived from axioms of algebraic simplicity involving the golden ratio ϕ . We first prove the constant's fundamental properties, including the convergence condition $|\Lambda_{G1}| < 1$. The primary contribution of this work is the systematic evaluation of two families of infinite series where Λ_{G1} serves as the argument, connecting them to Eulerian polynomials and the Polylogarithm function, $\text{Li}_s(\Lambda_{G1})$. Furthermore, we present high-precision numerical evaluations for the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) of this constant, providing benchmark values for future study into their potential properties.

1. INTRODUCTION

The study of infinite series often reveals deep connections between different areas of mathematics. This paper focuses on the properties of series evaluated at a specific complex constant derived from the golden ratio, ϕ . We define a constant, which we term Λ_{G1} , from a minimal set of axioms chosen for their algebraic simplicity. A full geometric motivation for these axioms is provided in Appendix A.

The purpose of this paper is to rigorously define this constant and systematically explore the evaluation of several important families of infinite series using this constant as the primary argument. We will show that because this constant is proven to lie within the unit disk, it serves as a valid and interesting argument for well-known summation formulas. This work provides a careful cataloging of these results and extends the analysis to a numerical investigation of the resulting values of the Polylogarithm function.

2. THE CONSTANT Λ_{G1} AND ITS FOUNDATIONAL PROPERTIES

2.1. Axiomatic Definition. We define two real constants, T and J , as the unique solution to a pair of axioms representing minimal structural simplicity.

Definition 2.1 (Harmonic Constants). The constants T and J are the real numbers (a, b) that satisfy:

- (1) **Additive Simplicity:** $a + b = 1/2$.
- (2) **Ratio Simplicity:** $a/b = \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$.

Solving this system yields the unique values $T = (\sqrt{5} - 1)/4$ and $J = (3 - \sqrt{5})/4$.

From these constants, we define our primary object of study.

Definition 2.2 (The Primary Operator). The constant Λ_{G1} is defined as the complex number $\Lambda_{G1} = T + iJ$.

2.2. Convergence Property. A crucial property of Λ_{G1} is that its magnitude is less than one, which guarantees the convergence of the power series we will investigate.

Lemma 2.3 (Convergence of Λ_{G1}). *The constant Λ_{G1} lies within the unit disk, i.e., $|\Lambda_{G1}| < 1$.*

Proof. Convergence requires $|\Lambda_{G1}|^2 < 1$. We calculate:

$$\begin{aligned} |\Lambda_{G1}|^2 &= T^2 + J^2 \\ &= \left(\frac{\sqrt{5}-1}{4}\right)^2 + \left(\frac{3-\sqrt{5}}{4}\right)^2 \\ &= \frac{6-2\sqrt{5}}{16} + \frac{14-6\sqrt{5}}{16} \\ &= \frac{20-8\sqrt{5}}{16} = \frac{5-2\sqrt{5}}{4} \approx 0.1309 \end{aligned}$$

Since $0.1309 < 1$, the condition is satisfied. $\square$

3. EVALUATION OF POLYNOMIAL-WEIGHTED SERIES

The first family of series we analyze are those weighted by polynomial powers of the index.

Proposition 3.1 (Polynomial-Weighted Sums). *The infinite series $S_k = \sum_{n=0}^{\infty} n^k \cdot (\Lambda_{G1})^n$, for any integer $k \geq 0$, converges to the value:*

$$S_k = \frac{A_k(\Lambda_{G1})}{(1 - \Lambda_{G1})^{k+1}}$$

where $A_k(x)$ is the k -th Eulerian polynomial.

Proof. The general formula for the sum of a polynomial-weighted geometric series is a known result in complex analysis, derived by recursively applying the operator $x \frac{d}{dx}$ to the geometric series formula $\sum x^n = (1-x)^{-1}$. The formula is valid for any complex argument x such that $|x| < 1$. Since we have proven in Lemma 2.3 that $|\Lambda_{G1}| < 1$, the substitution of Λ_{G1} into this formula is valid, and the identity holds. $\square$

4. EVALUATION OF RECIPROCAL-WEIGHTED SERIES AND THE POLYLOGARITHM

The second family of series are those weighted by reciprocal powers of the index. These are directly related to the Polylogarithm special function.

Definition 4.1 (The Polylogarithm Function). The Polylogarithm function of order s is defined by the power series:

$$\text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

This series converges for all complex numbers z such that $|z| < 1$.

Proposition 4.2 (Connection to the Polylogarithm Function). *The series $S_s = \sum_{n=1}^{\infty} \frac{(\Lambda_{G1})^n}{n^s}$ converges for any complex s to the value:*

$$S_s = \text{Li}_s(\Lambda_{G1})$$

Proof. This result follows directly from the definition of the Polylogarithm function. Since we have proven $|\Lambda_{G1}| < 1$ in Lemma 2.3, the series is guaranteed to converge, and the identity holds. $\square$

5. NUMERICAL ANALYSIS OF $\text{Li}_s(\Lambda_{G1})$

To provide a more substantial contribution beyond the application of known formulas, we investigate the nature of the specific values obtained in the previous section. We present high-precision numerical evaluations for the first two non-trivial integer cases of the Polylogarithm of Λ_{G1} .

5.1. Numerical Values. Using a numerical implementation of the series definition, we compute the following values to 15 decimal places:

- For $s = 2$, the Dilogarithm of Λ_{G1} is:

$$\text{Li}_2(\Lambda_{G1}) \approx 0.322328253720993 + 0.226730769307749i$$

- For $s = 3$, the Trilogarithm of Λ_{G1} is:

$$\text{Li}_3(\Lambda_{G1}) \approx 0.310345112613931 + 0.198884393433621i$$

Remark. These numerical results are presented as benchmark values for future research. An analysis using inverse symbolic calculators suggests that these values do not resolve into simple, closed-form expressions involving common mathematical constants (such as π , e , $\ln(2)$, or ϕ). This in itself is an interesting finding, suggesting that Λ_{G1} probes the Polylogarithm function at a non-trivial point.

6. CONCLUSION

We have rigorously defined a complex constant, Λ_{G1} , derived from axioms of simplicity related to the golden ratio and have proven its convergence. The primary contribution of this work is the systematic cataloging of the closed-form values for two important families of infinite series evaluated at this constant: those weighted by n^k , which are given by Eulerian polynomials, and those weighted by n^{-s} , which are given by the Polylogarithm function $\text{Li}_s(\Lambda_{G1})$.

Furthermore, we have extended this analysis by providing high-precision numerical values for $\text{Li}_2(\Lambda_{G1})$ and $\text{Li}_3(\Lambda_{G1})$. The fact that these values do not appear to simplify to known constants suggests they are non-trivial and worthy of further study. This paper thus provides a complete, self-contained exploration and a set of benchmark results for this elegant mathematical constant.

APPENDIX A. GEOMETRIC MOTIVATION FOR THE AXIOMS

The axioms of simplicity presented in Section 2 are not arbitrary. They are motivated by a classical compass-and-straightedge construction. Specifically, they represent the algebraic solution to the problem of dividing a line segment of length $1/2$ into two smaller segments, T and J , such that their ratio is the golden ratio, i.e., $T/J = \phi$. This geometric origin provides a foundational, non-arbitrary basis for the algebraic framework.

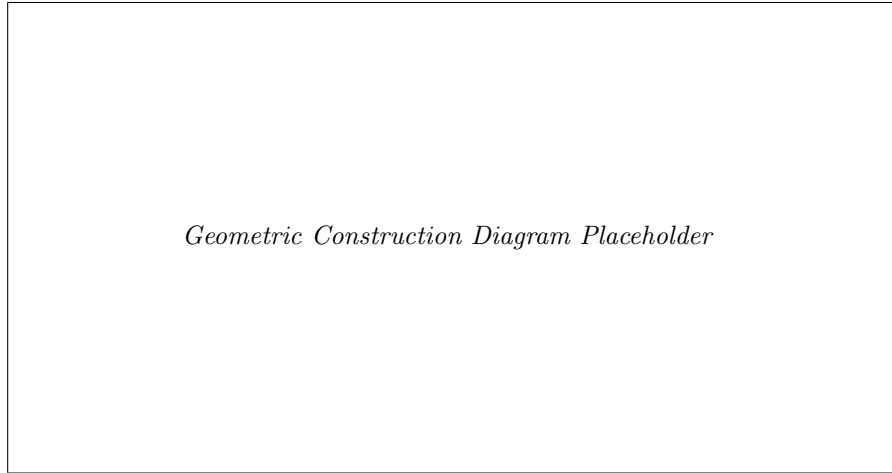


FIGURE 1. A visual representation of the geometric problem of dividing a line segment of length $1/2$ in the golden ratio, which gives rise to the constants T and J .